Basic Sight-Singing and
Ear Training for Church Singers

Unit 10.2 - Pitch

Basic Harmony –

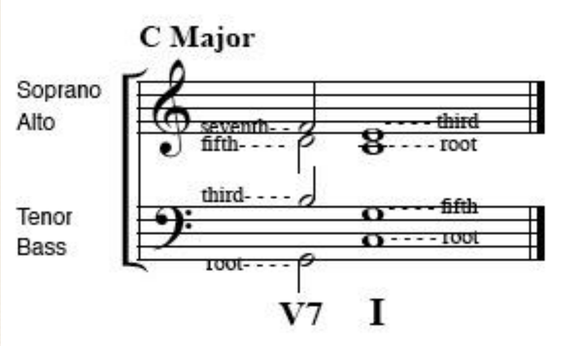
The Dominant 7th Chord.
Chord Progressions.
The Hierarchy of Choral Voices.

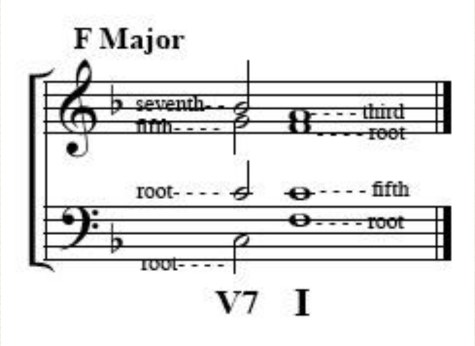
The Dominant 7th Chord

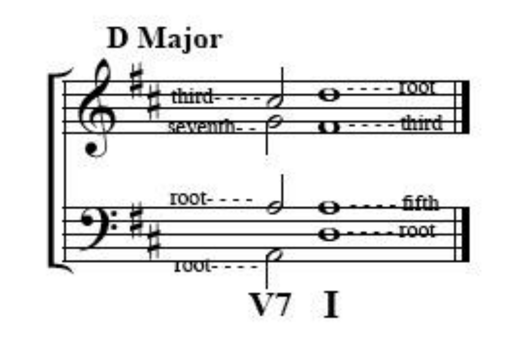
The principle of constructing chords by stacking thirds on top of a root note can be extended beyond beyond the **triad** (which by definition consists of three notes). If another (minor) third is added on top of a major triad, the result is a **seventh** chord, so called because the resulting interval from the bottom note to the top is a **seventh**.

In order to construct such a chord using only the notes of the major scale, it needs to be built upon the **V degree** (the fifth degree) of the scale, known in musical terminology as the **dominant**. This chord is known as the **dominant 7th chord**. In Western tonal music (which includes most Orthodox choral music written in four parts) the dominant 7th chord is perceived as having a certain degree of **tension** (that is to say, **dissonance**), which requires **resolution** achieved by moving to the chord built on the **I degree** (the first degree)—the **tonic**. In fact, the dominant 7th chord typically contains the dissonant interval of the **tritone**—between the **third** and the **seventh** note of the chord.

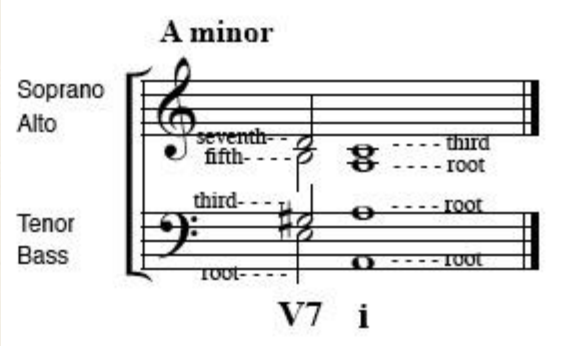
Here are some common examples of dominant 7th chords and their resolutions in various keys:

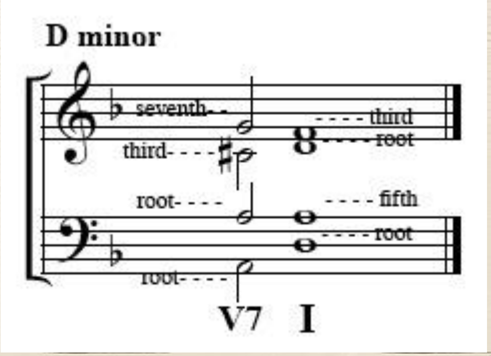


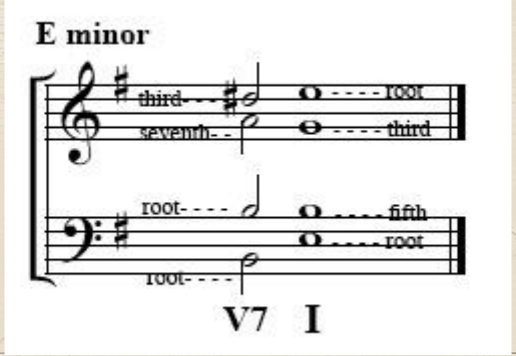




In minor keys, the dominant 7th chord always involves a raised seventh scale degree (part of the **harmonic minor scale**), which is why dominant 7th chords in minor keys contain sharps as accidentals.





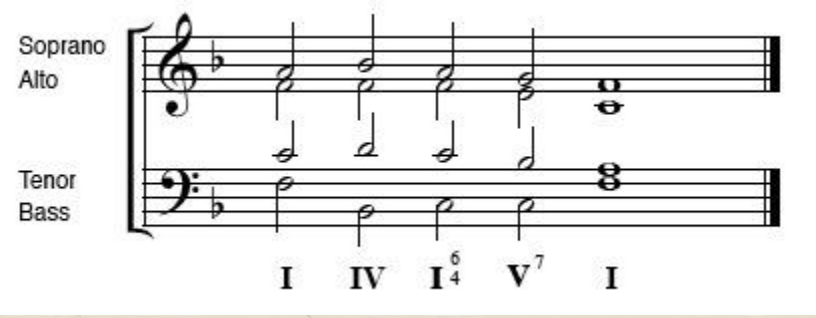


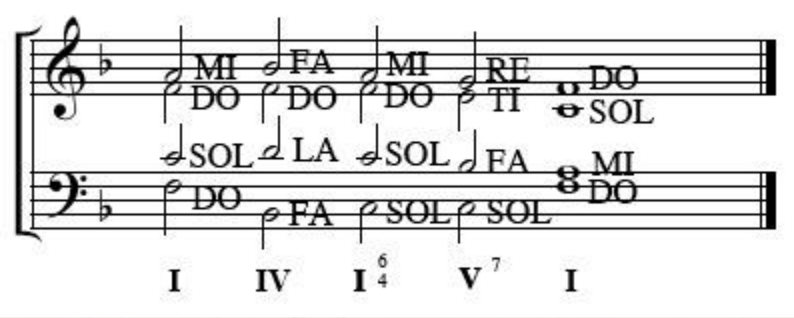
The psychological (and acoustical) effect of tension building and resolving has the effect of propelling harmonic music forward, from a starting point (often the tonic), leading **away** from the tonic (in effect, building up tension), and finally **resolving** the tension on the tonic, which creates in the listener a sense of **arrival** (return) and **satisfaction** (repose). This can happen both within several chords (on the "micro" level) as well as within an entire extended piece of music or section (movement) thereof (on the "macro" level).

To experience this effect, all one has to do is to reverse the order of the chords from the previous slide—moving from the I to the V7 chord. If a piece of music or a hymn were to end on the 7th chord, one would experience a profound sense of "incompleteness" and "discomfort" (dissatisfaction), in comparison to the "satisfaction" one feels when hearing the dominant 7th moving to the tonic.

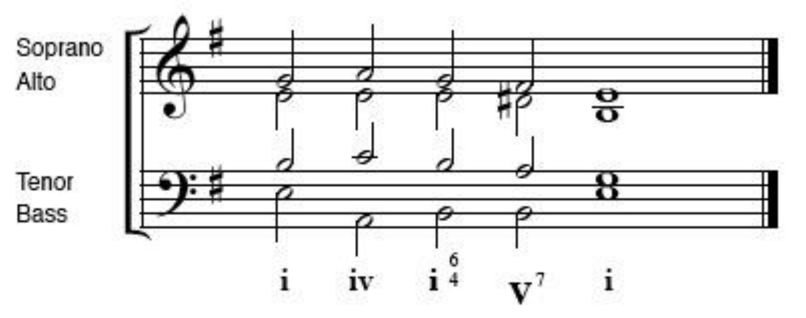
Chord Progressions

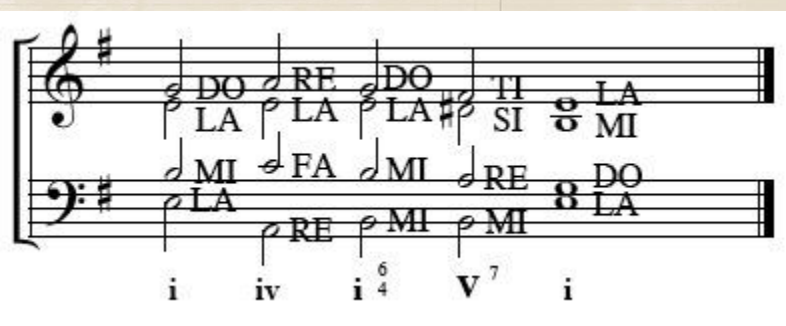
An orderly sequence of chords following the rules of harmony, which leads the ear of the listener forward, is called a **chord progression**. The simplest types of chord progressions, which begin on the **tonic**, move away to the **subdominant** (the chord built on the IV scale degree, the note below the dominant), then to the **dominant** (with or without the seventh), and then back to a resolution on the tonic, are well familiar to the ears of most Orthodox choral singers. In musical shorthand, such a chord progression is expressed as I–IV–V–I (or I–IV–V^7–I). Here is this chord progression in the key of F major. Note that all the chord tones can be expressed in terms of SOL-FA syllables appropriate to the given key:



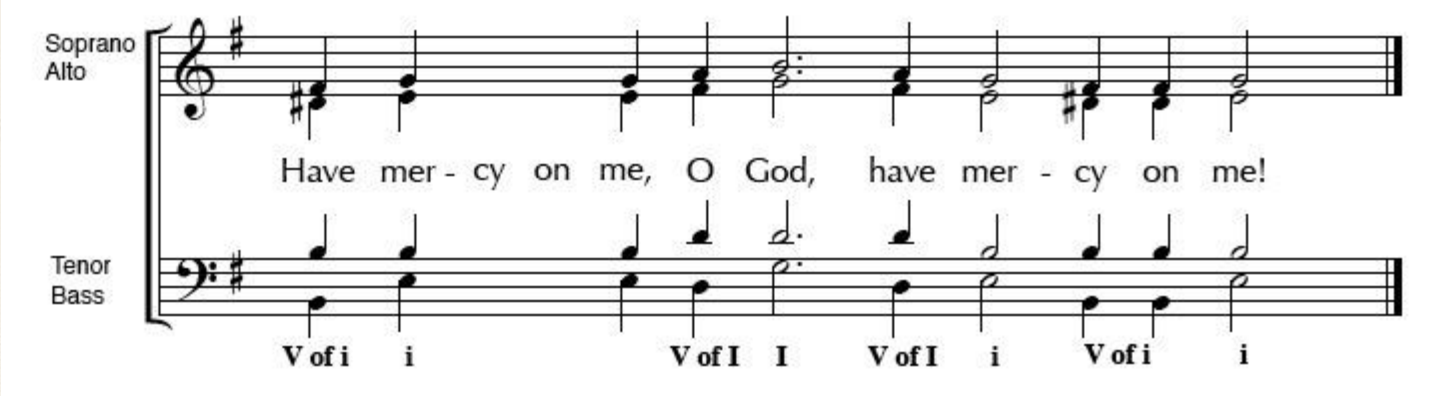


In a minor key, such as E minor, the shorthand (using small Roman numerals for minor chords) for a frequently heard chord progression is: i–iv–$i^{6/4}$–V^7–i.





When a composer or arranger takes a melody, for example, a pre-existing chant, and arranges it in four-part harmony, the result is referred to as a **harmonization**; in essence, the melody is supplied with chords that form a **chord progression**. Certain kinds of chord progressions, such as those found in 19th-century harmonizations of Kievan Chant, appear to pivot between a **minor tonic** (shown by the lower-case Roman numeral **i**) and a **major tonic**—the relative major (shown by the upper-case Roman numeral **I**), as shown in the example on the next slide.



This type of very simple **harmonic analysis**—the awareness of how in church music chords supplant each other in familiar and often predictable patterns—is very helpful to the process of sight reading.

The study of chord progressions and how they are formed through the horizontal movement of different voices is a fascinating and complex topic, one that typically may involve a year or two of focused study—the subjects of Harmony and Counterpoint—on the college or university level. The introductory information presented here is intended to heighten singers' awareness of the two dimensions in which they participate—the **horizontal** (melodic) and the **vertical** (harmonic).

A Choral Singer's "Harmonic Awareness"

Orthodox choral music written for four voices offers considerable variety of possible combinations of melody and harmony. Within this variety, however, certain structural principles can be observed. The skilled choral singer needs to be aware of these structures, insofar as they directly affect the singer's task. For example, at any given time, when singing a melody that forms one of the constituent notes of a chord, the well-qualified choir singer must be able to answer the following questions:

- Am I singing an important part (i.e., the melody) or a subordinate part?
- What is the relative importance of my part in the overall choral fabric?
- Is there another part with which I am singing in parallel motion?
- Are all the parts moving in the same rhythm (homorhythmically) or is my part more or less rhythmically active than the others?
- Is the note I am singing in a given chord the root, the third, the fifth (or perhaps, the seventh)?

- Is the chord major or minor, and am I singing the part that determines which one it is (that is, the third of the chord)?
- Is another voice part singing the same chord tone that I am? Which one? Are we in tune with one another?

The "Hierarchy" of Choral Voices

As the questions above imply, in harmonized choral music we can identify some voices as being more important that others. Certainly this is the case in chant harmonizations, where a pre-existing **chant melody** has had other parts added to it, forming chords. As mentioned earlier, this process occurred chiefly in the late 17th and early 18th centuries in Ukraine and Russia. From there, immigrants brought harmonized chant to America, where it has been hospitably received by worshipers already acclimated to Western music.

The earliest chant harmonizations were modeled upon popular para-liturgical songs (similar to carols) called **kants** (from the Latin "*cantus*"—a song). These sacred songs were popular in Poland and other Roman Catholic lands, from where they made their way to Ukraine and then to Russia. The typical kant consisted of three voice parts: two voices singing the melody in **parallel thirds**, and a bass part that jumped back and forth, singing the roots of the chords suggested by the movements of the melodic voices. The following excerpt from a Lenten hymn illustrates the structure of the kant style:

Notice that the harmony here is identical to the example on Slide 7, with a minor tonic alternating with a major tonic.

The same principles of harmonization were applied during the 18th century to numerous common (**obihod**) chants of the Eight-Tone cycle. Initially, singers improvised the harmonies—only the melody and the bass were actually written down. In the mid-19th century, the fully harmonized, 4-part versions of the chants were written down and published by the Russian Imperial Chapel. The so-called “Obihod” harmonizations proved to be very enduring, because they could easily be sung by choirs of any size in a variety of voicings: unison, 2-part, 3-part, 4-part, and even 6- or 8-part.

The following well-known example, in Common Russian Tone 2, illustrates the “hierarchy” of voice parts in common Russian chant.

One possible harmonization keeps the melody in the top voice, the **soprano**, and adds harmony a sixth below in the **tenor**. The **bass** sings the roots of the chords suggested by the two upper voices, and the **alto**—the “filler”— fills in the missing chord tones of the triads.

A different harmonization scheme puts the melody in the alto and the harmony in the soprano. The tenor becomes the filler.

The choral singer who is **aware** which part he or she is singing will make sure:

1. That the **melody** is prominent, even slightly more prominent than the harmony.
2. That the two parts singing the melody and the harmony are **working together rhythmically**.
3. That the **filler** part is not too loud.
4. That all parts can hear the bass, since the triads are all built upon the bass as a solid acoustic “**foundation.**”
5. That whoever is singing the F# accidental, which is the “leading tone” to the tonic G minor, is singing their part **extra high**, keeping the major triad in tune and preventing the overall pitch from sagging.

This type of sample analysis and its practical implementation must be done **by the choir director** with the unfailing assistance and participation of **aware and educated** choir singers.

Church choral singing performed in this way—without any annoying musical deficiencies or technical shortcomings, becomes the **ideal vehicle** for liturgical prayer.

Pitch Exercises 10.2 are intended as a summary and a reference for future use.

C Major
G Major
F Major
D Major

A minor
E minor
D minor
G minor

INSTRUCTIONS:
Use each solfège table to review the scales and intervals in each key.
Then, depending on your voice part, practice sight-singing the Typical Patterns for your voice.

Congratulations! You have taken your **initial steps** to becoming an aware and educated choir singer!

This process does not happen immediately, certainly not after taking a ten-week online course.

Please take care to **preserve**, **develop**, and **cultivate** what you have learned here!